

Exercise 1: Sample Bias

Assume that you contact a sample of people with different score of extraversion. Assume that these scores range on a scale from 0% to 100%. Moreover, a person's probability to actually respond to the survey invitation and report their extraversion score is this score itself.

- a) In any given population/sample, what is the expected actual score?
- b) In contrast to this, which average score can you expect to find?

Exercise 2: AI Checks

Assume that you include a control question in your survey to make sure to identify AI bots to exclude them from your data. Specifically, you use the ARC-style question below, of which you assume that $p_h = 95\%$ of human participants will actually select the correct answer (i.e., "B") and that AI bots will not do better than chance (i.e., randomly picking the correct answer in $p_c = 25\%$ of all cases). Looking at your data, you see that $C = 70\%$ of the sample has answered the control question correctly.

- a) What is your estimate for the share of AI bots in your sample?
- b) Plot this result for different values of p_h (x-axis from 0 to 100%) and C (40%, 50%, ..., 90% as color).

Exercise 3: Constructs and Items

You want to conduct a survey and assess your participants' perceptions of *flow* (i.e., the mental state in which a person is fully immersed in an activity, experiencing energized focus, deep involvement, and enjoyment in the process).

- a) Conduct a literature search on Google Scholar (or other tools) to identify a couple of established papers for this regard to identify a scale.
- b) Use ChatGPT (or any other AI chat tool) to create 3 to 6 items yourself.

Exercise 4: Networks and Causality

You may have heard the conventional wisdom saying that *"you are the average of the five people you spend the most time with"*. Implicitly, this is sometimes assumed to imply that your closest friends and peers have an effect on you, your personality, your success in life etc.

- a) But can this causal claim actually be made?
- b) What other possibilities are there that would provide an explanation for any observed correlation?

Exercise 5: Cronbach's Alpha & Explorative Analysis

In your survey, you have employed a multi-item construct with four items. Five participants took part in the survey.

- a) Based on their responses (below), what is the construct's Cronbach's alpha?

	Item 1	Item 2	Item 3	Item 4
Participant 1	1	5	4	3
Participant 2	2	4	3	4
Participant 3	2	2	2	2
Participant 4	3	4	4	3
Participant 5	2	5	5	4

The file *survey_data.csv* contains data from a survey with 8 constructs and a total of 31 items.

- b) What are the constructs' Cronbach's alpha values?
- c) Use a correlation plot to visualize the relations between all 31 items.
- d) For each construct, generate a new column with the mean of all of its items. Visualize the correlations between these aggregated construct values (e.g., correlation matrix with `ggally`).